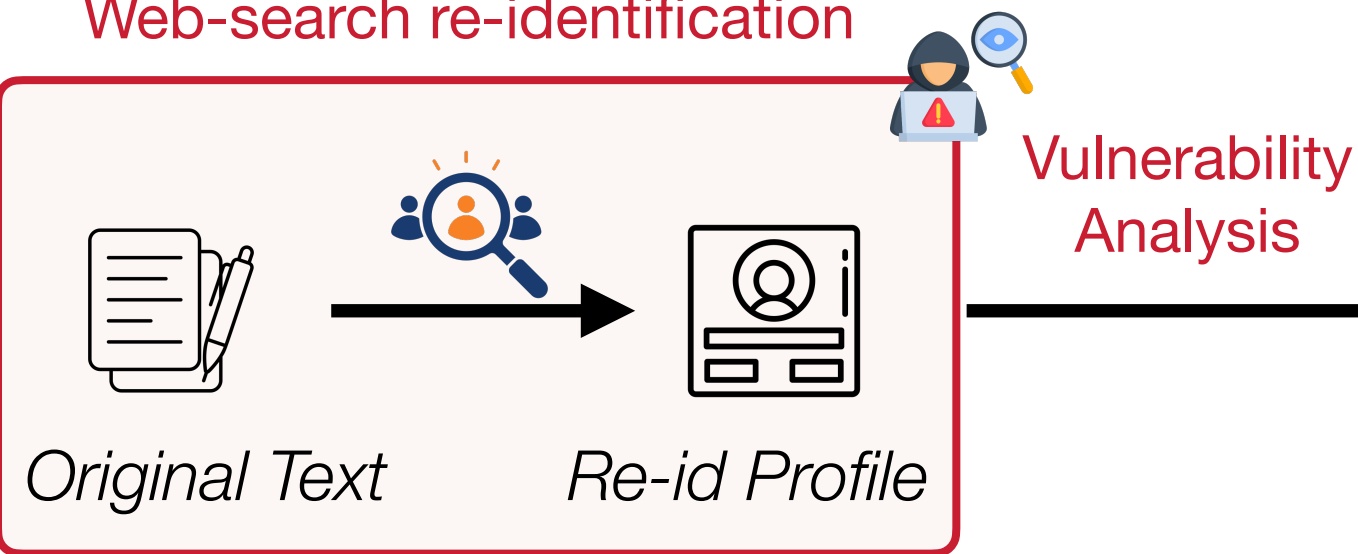


Adaptive Privacy Scope Expansion

Web-search re-identification



Transcript-specific quasi-identifiers

Workflow cues
Research pipeline
Tool-stack cues
Domain-practice cues
Institutional/contextual clue

Examples

I work in applied sensor physics, specifically on detecting weak environmental fields using tabletop interferometry... We wrote a paper, currently in review, which proposed a generalised mechanism for modelling the noise produced by moving calibration objects.

Before

I work in **scientific research that involves studying noise from environmental vibrations on sensitive measurement equipment...** We **completed** a **project** that proposed a **general approach** for modeling **the** noise **from** moving **sources**.

After

Original Qualitative Texts

I work in applied sensor physics, specifically on detecting weak environmental fields using tabletop interferometry... We wrote a paper, currently in review, which proposed a generalised mechanism for modelling the noise produced by moving calibration objects.

Phase 0: Initialization

Privacy Inference

Infer attributes using web-search re-identification inference attack

Utility Profiling

Extract research-valuable insight profile

Adaptive Privacy Attributes **A**

Base 8 Attributes

+

Transcript-specific

Insight profile **P**

Phase 1: Masking Convergence

Refresh privacy inference

Rewrite whole transcript

Repeat for R rounds

Diff original vs. rewritten transcript

Mask template $\hat{T}$ with $[MASK_i]$ + mask map M

Phase 2: Reconstruct, evaluate, select

Generate N candidate rewrites for masked spans

Attacker:

Privacy severity **S**
+ too specific count **C**

Keeper:

Utility loss **L**

Select candidate satisfying $C \leq C_{max}$, then minimize privacy severity **S** and utility loss **L**

Final sanitized transcript **T***